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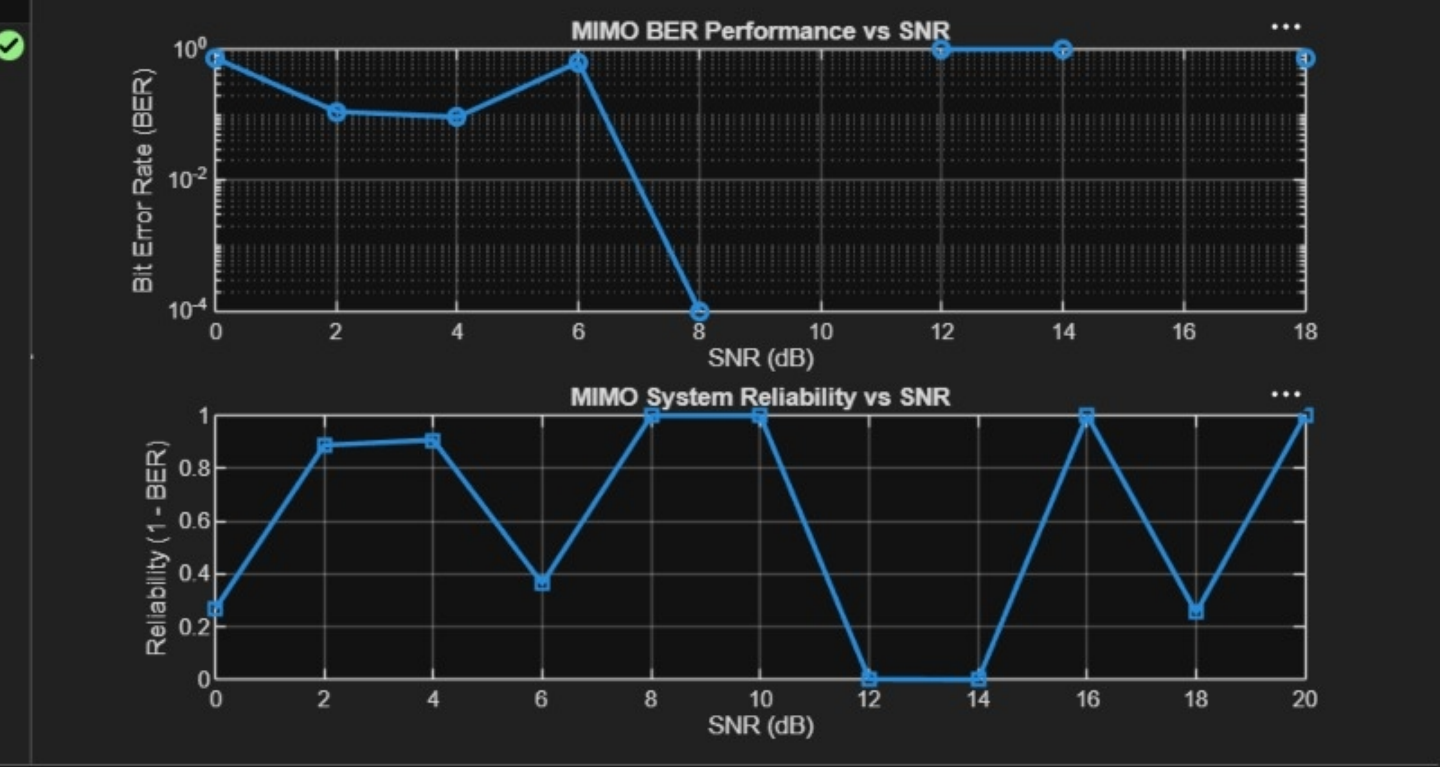
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 Legend
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 X-Grid
 Y-Grid

FILE LABELS AND ANNOTATIONS

```

12
13 H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
14 noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
15
16 tx = H(1,1)*symbols' + noise(1,:);
17 rx = real(tx) > 0;
18
19 BER(i) = sum(rx' ~= bits)/length(bits);
20
21 end
22 figure
23 % ----- Graph 1: BER vs SNR -----
24 subplot(2,1,1)
25 semilogy(SNR_dB,BER,'-o','LineWidth',2)
26 title('MIMO BER Performance vs SNR')
27 xlabel('SNR (dB)')
28 ylabel('Bit Error Rate (BER)')
29 grid on
30 % ----- Graph 2: SNR vs Quality Indicator -----
31 subplot(2,1,2)
32 plot(SNR_dB,1-BER,'-s','LineWidth',2)
33 title('MIMO System Reliability vs SNR')
34 xlabel('SNR (dB)')
35 ylabel('Reliability (1 - BER)')

```



```

>>

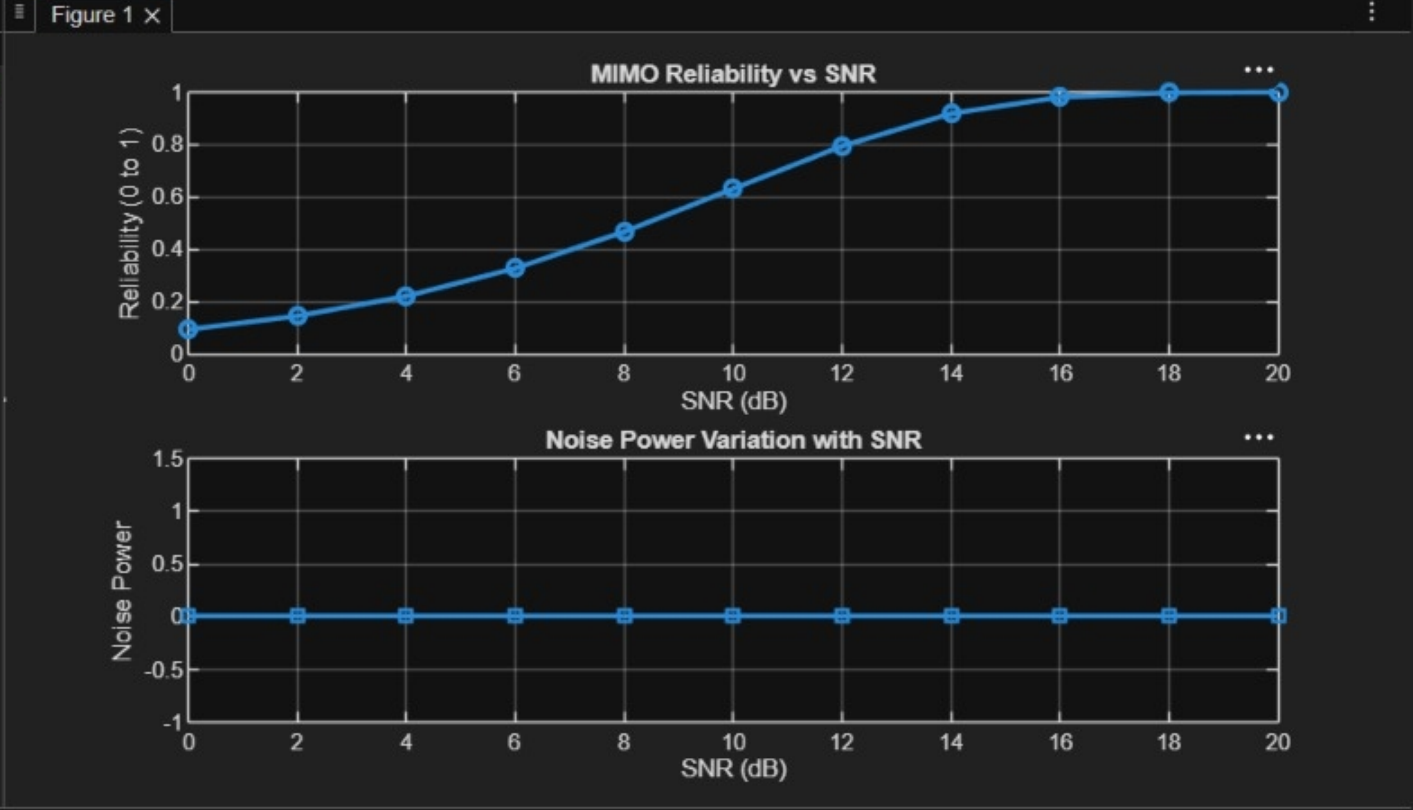
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LABELS AND ANNOTATIONS

```
9 for i = 1:length(SNR)
10     % simple channel model effect
11     H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
12
13     noise_power = 1/SNR(i);
14
15     % reliability increases with SNR
16     Reliability(i) = 1 - exp(-SNR(i)/10);
17 end
18 figure
19 % ----- Graph 1: SNR vs Reliability -----
20 subplot(2,1,1)
21 plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
22 title('MIMO Reliability vs SNR')
23 xlabel('SNR (dB)')
24 ylabel('Reliability (0 to 1)')
25 grid on
26 % ----- Graph 2: SNR vs Noise Effect -----
27 subplot(2,1,2)
28 plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)
29 title('Noise Power Variation with SNR')
30 xlabel('SNR (dB)')
31 ylabel('Noise Power')
```




```

7 alloc_RB = [10 12 8 15 5];
8 % Modulation efficiency (bits/symbol)
9 mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
10 % Throughput calculation (Mbps)
11 throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
12 % Display results
13 disp('User Throughput (Mbps):')
14 disp(throughput')
15 figure
16 % ----- Graph 1: Throughput per User -----
17 subplot(2,1,1)
18 plot(users, throughput, '-o','LineWidth',2)
19 title('OFDMA User Throughput')
20 xlabel('User Index')
21 ylabel('Throughput (Mbps)')
22 grid on
23 % ----- Graph 2: RB vs Throughput -----
24 subplot(2,1,2)
25 plot(alloc_RB, throughput, '-s','LineWidth',2)
26 title('Resource Blocks vs Throughput')
27 xlabel('Allocated Resource Blocks')
28 ylabel('Throughput (Mbps)')
29 grid on

```

The top plot, titled "OFDMA User Throughput", displays the throughput in Mbps for five different users. The x-axis represents the User Index (1 to 5), and the y-axis represents the Throughput (Mbps) from 0 to 10. The data points are connected by a blue line with circular markers.

User Index	Throughput (Mbps)
1	3.5
2	8.5
3	8.5
4	10.5
5	1.5

The bottom plot, titled "Resource Blocks vs Throughput", shows the throughput in Mbps for different numbers of allocated resource blocks. The x-axis represents the Allocated Resource Blocks (5 to 15), and the y-axis represents the Throughput (Mbps) from 0 to 10. The data points are connected by a blue line with square markers.

Allocated Resource Blocks	Throughput (Mbps)
5	1.5
8	8.5
10	3.5
12	8.5
15	10.5